

Part A – Setup & Dataset Loading

Q1. Import required libraries.

Q2. Load the dataset from your local path (after Kaggle download).

(<https://www.kaggle.com/datasets/nabeelqureshitiii/student-performance-dataset>)

Part B – Data Understanding

Q3. Display basic statistics and check for missing values.

Q4. Create a scatter plot showing how study hours affect total scores.

Q5. Visualize relationships between study hours, attendance, and total score. (Deep into)

Part C – Model Training (Linear Regression)

Q6. Define features (X) and target (y).

Q7. Split data into train and test sets.

Q8. Train a Linear Regression model.

Part D – Model Interpretation

Q9. Print the regression coefficients and intercept.

Q10. Evaluate the model using R^2 score.

Part E – Feature Importance (Tree-based Comparison)

Q11. Train a Random Forest model for comparison.

Q12. Extract and visualize feature importance.

Part F – Reflection Questions

Q13. Which feature has the highest impact in Linear Regression?

Q14. Which feature has the highest importance in Random Forest?

Q15. What's the main difference between Linear Regression and Random Forest interpretability?

Q16. Why is interpretability important in analytics?

Q17. Visualize actual vs predicted scores.

Part G – Model Complexity Comparison (Linear vs Random Forest)

Part H – Visualizing Overfitting vs Underfitting

Part I - Regularization Visualization (Ridge vs Lasso)